AI Playbook and Toolkit for Public Health Departments

AI for Public Health Communications and Message Development

COM 200

Generative AI can support public health communications by helping staff draft, revise, simplify, translate, tailor, and test messages. However, public health communication carries high trust, equity, accessibility, and accuracy obligations. AI-generated text can sound clear and authoritative while misstating evidence, omitting uncertainty, using stigmatizing language, failing to meet accessibility needs, or creating messages that do not fit community context. This module teaches communications staff and program partners how to use AI as a drafting and review aid while preserving human accountability. It focuses on public-facing messages, internal communications, plain language, community relevance, multilingual communication, misinformation risks, and review workflows. The goal is not to replace skilled communicators. The goal is to help communications teams use AI safely for efficiency, brainstorming, and consistency while retaining professional judgment, source verification, community review, and public trust.

Level	applied/foundational
Primary track	Communications Role-Based Track
Appears in tracks	communications, shared-foundational, policy, public-health-executive-leadership

Learning Objectives

- Identify appropriate and inappropriate uses of AI in public health communications.
- Apply source verification, plain language, accessibility, and equity review to AI-generated messages.
- Design a human review workflow for AI-assisted public-facing communications.
- Recognize misinformation, overstatement, missing uncertainty, and stigmatizing language risks.
- Produce an AI-assisted communications review checklist.

Module Overview

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This module teaches communications staff and program partners how to use AI as a drafting and review aid while preserving human accountability. It focuses on public-facing messages, internal communications, plain language, community relevance, multilingual communication, misinformation risks, and review workflows.

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Public Health Example

A health department uses AI to draft a heat safety message for older adults, outdoor workers, and people without reliable air conditioning. Communications staff verify the recommendations against official guidance, rewrite the message in plain language, review it for accessibility and language access, avoid blaming individuals for structural risks, and route the final version through the approved public information process before release.

Topic Deep Dive

AI can be useful early in the message development process. It can generate first drafts, propose alternative tones, simplify technical language, identify jargon, create audience-specific versions, and suggest questions the public may ask. These uses can save time, especially when communications staff support multiple programs. However, AI should not be treated as the authority on public health content. The approved evidence base should come from agency guidance, subject matter experts, official data, and community context.

Source verification is essential because AI-generated language may include unsupported claims, outdated recommendations, fabricated details, or overconfident statements. Communications staff should verify factual claims, numbers, dates, risk statements, recommendations, and eligibility criteria against approved sources. When uncertainty exists, the message should explain what is known, what is not known, and what actions people can take.

Equity and accessibility review should be built into the workflow. AI-generated messages may not adequately address language access, disability access, literacy, cultural context, rural access, transportation barriers, technology access, or historical mistrust. Public health messages should be reviewed for plain language, readability, alternative formats, translation quality, non-stigmatizing language, and practical feasibility for the intended audience.

AI can also support review, but it should not replace reviewers. A communications workflow should define who checks technical accuracy, who reviews equity and accessibility, who approves public release, and who monitors public response after dissemination. For urgent communications, the workflow should include expedited review rules that still preserve accuracy and accountability.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

AI-generated messages overstating evidence or omitting uncertainty.

Use of stigmatizing, blaming, or culturally inappropriate language.

Unverified translations or readability changes that alter meaning.

Public-facing content being released without required subject matter or leadership review.

Messages that assume access to resources, technology, transportation, or healthcare that the audience may not have.

Recommended guardrails include the following:

Use only approved tools and approved source material for public health communications.

Require human review for accuracy, plain language, accessibility, equity, and tone.

Document source material used for public-facing claims.

Use qualified review for translation and language access needs.

Route public-facing materials through the agency approval process before release.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Create an AI-assisted communications review checklist for one public health message. Include intended audience, source material, factual claims to verify, plain language review, accessibility review, language access review, equity review, approval pathway, and documentation expectations.

Expected Artifact or Evidence

AI-assisted communications review checklist

Source verification notes

Plain language and accessibility review notes

Approval pathway documentation

Final reviewed message or sample revision

Expected Assignment

- AI-assisted communications review checklist
- Source verification notes
- Plain language and accessibility review notes
- Approval pathway documentation
- Final reviewed message or sample revision

Knowledge Check

- 1. What is an appropriate role for AI in public health communications?
- 2. Why is source verification necessary?
- 3. Which review is especially important for public-facing AI-assisted messages?
- 4. What should happen with AI-assisted translation for public health messages?
- 5. What is strong completion evidence?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP guidance for LLM application security
- Agency policies for privacy, cybersecurity, records, procurement, accessibility, and public communications
- Relevant public health program SOPs, data governance documentation, and implementation plans